

MATH565: Combinatorics and Graph Theory

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Abstract

This is a template for TCS (and other kind of) lecture notes.

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Part I

Extremal Graph Theory

Chapter 1

Turán's Theorem

Question 1.0.1. How many connections does a group of n people we need to gather to ensure that there exists a triplet of mutual friends?

To formalize this question, we need to introduce the basic definition of graph theory.

1.1 Multisets and Combinatorial Counting

Definition 1.1.1 (Multiset). A **multiset** M on a set S is a collection (unordered) of elements in S . $\forall x \in M \cap S$, the number of times that $x \in S$ occurs in M is greater or equal to 0 and is called the **multiplicity** of x in M , $\mu_M(x)$.
If $\forall x \in S, \mu_M(x) = \mu_{M'}(x)$, then $M = M'$.

Example 1.1.1. If $\mu_M(x) = 0, 1 \quad \forall x \in S$, then M is a set.

Definition 1.1.2. Let S be a finite set of size n .

$$\binom{S}{k} = \{k\text{-subsets of } S\} = \{A \subseteq S : |A| = k\}$$

$$\left| \binom{S}{k} \right| = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\left(\binom{S}{k} \right) = \{k\text{-multisets of } S\} = \{M : M \text{ is a multiset on } S, |M| = k\}$$

$$\left| \left(\binom{S}{k} \right) \right| = \binom{n+k-1}{k}$$

There're 2 ways to compute the size of $\left(\binom{S}{k} \right)$.

Assume a grid of size $n \times (k+1)$. We label the columns with elements in S , and rows from 0 to k . The problem is then equivalent to finding the ways to walk from $(0, S_1)$ to (k, S_n) , where we can only move right or up (the number of steps we go up is the multiplicity of the corresponding element in the multiset). The number of such paths is $\binom{n+k-1}{k}$.

Alternatively, we know that $\sum_{i=1}^n \mu_M(S_i) = k$, and $\mu_M(S_i) \geq 0$. This is equivalent to finding the number of non-negative integer solutions to the equation $x_1 + x_2 + \dots + x_n = k$, which is a classic combinatorial problem. Using stars and bars, we can find that the number of solutions is $\binom{n+k-1}{k}$.

1.2 Graph

Definition 1.2.1 (Finite Graph). A **finite graph** G is a triple (V, E, φ) , where

- $V = \{v_1, v_2, \dots, v_n\}$,
- $E = \{e_1, e_2, \dots, e_q\}$,
- φ is a function from E to $\binom{V}{2}$.

Definition 1.2.2 (Simple Graph). A finite graph is **simple** if φ is injective (i.e. no multi-edges) and $\text{Im}(\varphi) \subseteq \binom{V}{2}$ (i.e. no loops).

For the sake of simplicity, we use the following notations from now on:

- $\varphi(e) = (u, v)$ is denoted as $e = (u, v)$;
- If $e = (u, u)$, we say e is a loop;
- Subset I of V is independent if no pairs in I are in E .

Definition 1.2.3 (Complete Graph and Complete Bipartite Graph). For $n \geq 3$, V with $|V| = n$, the **complete graph** K_n is the graph $(V, \binom{V}{2}, \varphi)$, where φ is the isomorphism from $\{e_1, e_2, \dots, e_{\binom{n}{2}}\}$ to $\binom{V}{2}$.

A **complete bipartite graph** $K_{m,n}$ is a graph $(L \cup R, L \times R)$, where $|L| = m$, $|R| = n$ and $L \cap R = \emptyset$.

With such definitions, we can formalize Question 1.0.1 as follows:

Question 1.2.1. What is the minimum number of edges in a simple graph with n vertices that guarantees the existence of a K_3 subgraph?

Equivalently, we can ask the following question:

Question 1.2.2. What is the maximum number of edges in a simple graph with n vertices that does not contain a K_3 subgraph?

Think of a Complete Bipartite Graph $K_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor}$, which has $\lfloor n^2/4 \rfloor$ edges and does not contain a K_3 subgraph. We can say the lower bound of this question is $\lfloor n^2/4 \rfloor$.

Actually, as Turán and Mantel's Theorem states, this is also the upper bound.

1.3 Turán and Mantel's Theorem

Definition 1.3.1 (Turán's Number). Let $n \geq 3$. The **Turán's number** $T(n) = \max\{|E(G)| : |V(G)| = n \text{ and } G \text{ does not contain a } K_3 \text{ subgraph}\}$.

Theorem 1.3.1 (Turán and Mantel's Theorem). For $n \geq 3$, $T(n) = \lfloor n^2/4 \rfloor$.

Proof. It suffices to prove that $T(n) \leq n^2/4$, i.e. for any graph G with no triangle subgraph on n vertices, $|E(G)| \leq n^2/4$.

Let $x \in V(G)$ be a vertex with maximum degree $\deg(x) = d$. From definition we know $N(x)$ is an independent set, and $|N(x)| = d$. Therefore,

$$|E(G)| \leq \sum_{v \notin N(x)} \deg(v) \leq \sum_{v \notin N(x)} d = (n - d)d \leq \max_d (n - d)d = n^2/4.$$

■

1.4 Extending to K_p

Similar to how we come up with bipartites, for K_p , we can consider $(p-1)$ -partite graphs. Formally, consider $K(n, p)$ as a multi-partite graph with:

$$n = t(p-1) + r, \quad 0 \leq r < p-1,$$

where t is the size of each partite set and r is the number of partite sets with size $t+1$. Then we have

$$V = S_1 \sqcup S_2 \sqcup \cdots \sqcup S_r \sqcup S_{r+1} \sqcup \cdots \sqcup S_{p-1},$$

where $|S_i| = t+1$ for $i \leq r$ and $|S_i| = t$ for $i > r$ (the symbol $\sqcup$ denotes disjoint union, a union of sets that are pairwise disjoint). The edge set in $K(n, p)$ is

$$E = \{(v, u) : v \in S_i, u \in S_j, i \neq j\}.$$

Calculating the number of edges in $K(n, p)$ (we omit the details here), we have

$$M(n, p) = |E(K(n, p))| = \frac{(p-2)n^2}{2(p-1)} + \frac{r(p-1-r)}{2(p-1)}.$$

Theorem 1.4.1 (Turán's Theorem). If a simple graph on n vertices contains more than $M(n, p)$ edges, then it contains a K_p subgraph.

Proof. We prove by induction on t . (Equivalently, we can prove by strong induction on n .)

Base case: $t = 0$. Then a clique K_n will never contain a K_p subgraph as $n < p$.

Inductive case: Assume the theorem holds for $t-1$.

Consider a graph G on n vertices with maximum number of edges so that it does not contain a K_p subgraph. We first claim G must contain a K_{p-1} subgraph. Otherwise, we can add an edge to G without creating a K_p subgraph, contradicting the maximality of G . Fix one K_{p-1} subgraph H . $|E(H)| = \binom{p-1}{2}$.

Let $G' = G \setminus H$ be the graph obtained by removing H from G . Then G' has $n - (p-1)$ vertices and does not contain a K_p subgraph. By the inductive hypothesis, we have

$$E(G') \leq M(n - (p-1), p).$$

For edges between H and G' , we claim that each vertex in G' can be adjacent to at most $p-2$ vertices in H . Otherwise, if a vertex $v \in G'$ is adjacent to $p-1$ vertices in H , then together with v and those $p-1$ vertices, we have a K_p subgraph. Therefore:

$$E(G) \leq E(G') + E(H) + (p-2)(n - (p-1)) \leq M(n - (p-1), p) + \binom{p-1}{2} + (p-2)(n - (p-1)) = M(n, p).$$

■

Chapter 2

Ramsey's Theorem

Question 2.0.1. How many people do we need to invite to a party so that there are either 3 people who all know each other or 3 people who are all strangers to each other?

2.1 Formalization

Definition 2.1.1 (Monochromatic Clique). Given a blue-orange coloring on $G = K_n$,

$$\beta(G) = \max\{S \subseteq V(G) : \text{all edges in } S \text{ are blue}\}$$

$$\omega(G) = \max\{S \subseteq V(G) : \text{all edges in } S \text{ are orange}\}$$

Any $S \subseteq V(G)$ such that all edges in S are the same color is called a **monochromatic** clique.

Theorem 2.1.1 (Ramsey's Theorem). For any blue-orange coloring on K_n , with $n \geq 6$, either $\beta(G) \geq 3$ or $\omega(G) \geq 3$.

Proof. Let $n \geq 6$, pick $x \in K_n$. By Pigeonhole Principle, either x is adjacent to at least 3 vertices in blue or at least 3 vertices in orange. WLOG, assume x is adjacent to at least 3 vertices in blue, say y_1, y_2, y_3 . If any edge between y_i and y_j is blue, then we have a blue triangle. Otherwise, all edges between y_i and y_j are orange, then we have an orange triangle. ■

Also, in K_5 , we can color the cycle C_5 in blue and the remaining edges in orange, so that there is no monochromatic triangle. Therefore, $n = 6$ is the minimum number of vertices needed to guarantee a monochromatic triangle.

2.2 Generalization

Definition 2.2.1 (Ramsey Number). For any $k, l \in \mathbb{Z}^+$, the Ramsey number $R(k, l)$ is the minimum number of vertices n such that for any blue-orange coloring on K_n , either $\beta(G) \geq k$ or $\omega(G) \geq l$.

Theorem 2.2.1 (Generalized Ramsey's Theorem).

$$R(k, l) \leq \binom{k+l-2}{k-1}$$

Note. The exact value of $R(k, l)$ remains open. What we know, for now, is that $R(4, 4) = 18$, $R(4, 5) = 25$. Even $R(5, 5)$ is still open.

Proof. We prove by induction on $k + l$.

Base case: when $k = 1$ or $l = 1$, for all $n \geq 1$, either $\beta(G) \geq k$ or $\omega(G) \geq l$ holds.

Inductive case: assume the theorem holds for all $k' + l' = k + l - 1$. Fix a blue-orange coloring on K_n . To prove $n \geq \binom{k+l-2}{k-1}$, it suffices to prove the case for $n = \binom{k+l-2}{k-1}$. Note that

$$\binom{k+l-2}{k-1} = \binom{k+l-3}{k-2} + \binom{k+l-3}{k-1}.$$

Fix a vertex $v \in K_n$. Let $O = \{u \in K_n : (u, v) \text{ is orange}\}$ and $B = \{u \in K_n : (u, v) \text{ is blue}\}$. Then $|O| + |B| = n - 1 = \binom{k+l-2}{k-1} - 1 = \binom{k+l-3}{k-2} + \binom{k+l-3}{k-1} - 1$.

We claim, either $|O| \geq \binom{k+l-3}{k-1}$ or $|B| \geq \binom{k+l-3}{k-2}$. Otherwise, we have

$$|O| \leq \binom{k+l-3}{k-1} - 1 \text{ and } |B| \leq \binom{k+l-3}{k-2} - 1 \Rightarrow |O| + |B| \leq n - 2 \Rightarrow \text{contradiction}.$$

- If $|O| \geq \binom{k+l-3}{k-1}$, then by the inductive hypothesis, either $\beta(O) \geq k$ or $\omega(O) \geq l - 1$. If $\beta(O) \geq k$, then we have $\beta(G) \geq k$. Otherwise, if $\omega(O) \geq l - 1$, then together with v , we have $\omega(G) \geq l$.
- If $|B| \geq \binom{k+l-3}{k-2}$, then by the inductive hypothesis, either $\beta(B) \geq k - 1$ or $\omega(B) \geq l$. If $\beta(B) \geq k - 1$, then together with v , we have $\beta(G) \geq k$. Otherwise, if $\omega(B) \geq l$, then we have $\omega(G) \geq l$.

In all cases, we have either $\beta(G) \geq k$ or $\omega(G) \geq l$. Therefore, $R(k, l) \leq n = \binom{k+l-2}{k-1}$. ■

The same strategy can be used to prove the following (we omit here):

Theorem 2.2.2.

$$R(k, l) \leq R(k, l - 1) + R(k - 1, l).$$

Part II

Graph Embeddings

Chapter 3

Planar Graphs

Question 3.0.1. When can a graph G be drawn in $\mathbb{R}^2$ with no edges crossing?

3.1 Formal Definitions

Definition 3.1.1 (Drawing of a graph). A **drawing** of a graph G is a representation of G in the plane $\mathbb{R}^2$ such that:

1. It's an injection from V to $\mathbb{R}^2$.
2. For each $e = (u, v) \in E$, an **arc** $\alpha_e \subseteq \mathbb{R}^2$ is drawn between $f(u)$ and $f(v)$ such that for all vertex w and edge e , $f(w)$ lies on α_e if and only if w is an endpoint of e .

Definition 3.1.2 (Planar (Embedding) and Faces). A drawing of a graph G is **planar** or is an **embedding** if no arcs meet except at shared endpoints, and no arcs meets itself.

The connected components of $\mathbb{R}^2 \setminus G$ for a planar graph are called **faces** of the drawing.

Theorem 3.1.1 (Jordan Curve Theorem). A Jordan curve $C \subset \mathbb{R}^2$ is a curve that can be continuously morphed to S^1 , the solution set of $x^2 + y^2 = 1$.

C divides the plane into two connected regions, one bounded and one unbounded.

Theorem 3.1.2. K_5 is not a planar graph.

Appendix

Appendix A

Algebra

Theorem A.0.1. For any polynomial $A \in \mathbb{C}[X]$ of degree $n - 1$, any point set $\{p_0, \dots, p_{n-1}\}$ of size n , $\{(p_i, A(p_i))\}_{i=0}^{n-1}$ determines $A(x)$ and vice versa.

Theorem A.0.2 (Fundamental Theorem of Algebra). Every non-constant single-variable polynomial with complex coefficients has at least one complex root.

In math terms, for $A(x) = \sum_{i=0}^{n-1} a_i x^i$, either $\exists x \in \mathbb{C}$ such that $A(x) = 0$, or $a_0 = a_1 = \dots = a_{n-1} = 0$.

Appendix B

Analysis

Theorem B.0.1 (Master Theorem (simplified)). Let the recurrence relation be:

$$T(n) = aT\left(\frac{n}{b}\right) + n^d$$

where $a \geq 1$ is the number of subproblems, $b > 1$ is the factor by which the problem size is reduced, and $d \geq 0$. The asymptotic bound of $T(n)$ is given by:

1. If $\log_b a > d$, then $T(n) = O(n^{\log_b a})$.
2. If $\log_b a = d$, then $T(n) = O(n^d \log n)$.
3. If $\log_b a < d$, then $T(n) = O(n^d)$.

Theorem B.0.2 (Master Theorem). Let $a \geq 1$ and $b > 1$ be constants, let $f(n)$ be a function, and let $T(n)$ be defined on the nonnegative integers by the recurrence:

$$T(n) = aT(n/b) + f(n)$$

Then $T(n)$ has the following asymptotic bounds:

- **Case 1:** If $f(n) = O(n^{\log_b a - \epsilon})$ for some constant $\epsilon > 0$, then $T(n) = \Theta(n^{\log_b a})$.
- **Case 2:** If $f(n) = \Theta(n^{\log_b a} \log^k n)$ for some constant $k \geq 0$, then $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$.
- **Case 3:** If $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some constant $\epsilon > 0$, and if $af(n/b) \leq cf(n)$ for some constant $c < 1$ and all sufficiently large n (Regularity Condition), then $T(n) = \Theta(f(n))$.